QUBO Formulation: Subset Sum Problem

Subset Sum Problem

Problem Statement. Given a set of N positive integers $\{a_1, a_2, \ldots, a_N\}$ and a target value T, the Subset Sum Problem seeks to identify a subset of these integers whose sum is as close as possible to T. Formally, the goal is to determine a selection of indices such that the absolute difference between the sum of the selected integers and the target value is minimized.

Decision Variables. We introduce binary decision variables $x_i \in \{0, 1\}$ for each $i \in \{1, \ldots, N\}$. The variable x_i indicates whether the i-th integer a_i is included in the subset:

$$x_i = \begin{cases} 1 & \text{if } a_i \text{ is selected,} \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

The sum of the selected integers is represented by $\sum_{i=1}^{N} a_i x_i$.

QUBO Derivation. The objective is to minimize the absolute deviation between the subset sum and the target. This can be expressed as:

$$\min \left| \sum_{i=1}^{N} a_i x_i - T \right|. \tag{2}$$

Since the square function is monotonically increasing for non-negative arguments, minimizing the absolute deviation is equivalent to minimizing the squared deviation:

$$\min \left(\sum_{i=1}^{N} a_i x_i - T \right)^2. \tag{3}$$

We expand the squared term algebraically:

$$\left(\sum_{i=1}^{N} a_i x_i - T \right)^2 = \left(\sum_{i=1}^{N} a_i x_i \right)^2 - 2T \sum_{i=1}^{N} a_i x_i + T^2 \tag{4}$$

$$= \sum_{i=1}^{N} \sum_{j=1}^{N} a_i a_j x_i x_j - 2T \sum_{i=1}^{N} a_i x_i + T^2. \tag{5}$$

Separating the terms where $i = j$ from those where $i \neq j$, and utilizing the idempotent property of binary variables $x_i^2 = x_i$, we obtain:

$$\sum_{i=1}^{N}\sum_{j=1}^{N} a_i a_j x_i x_j = \sum_{i=1}^{N} a_i^2 x_i^2 + \sum_{i\neq j} a_i a_j x_i x_j \tag{6}$$

$$= \sum_{i=1}^{N} a_i^2 x_i + \sum_{i\neq j} a_i a_j x_i x_j. \tag{7}$$

The double sum over $i \neq j$ can be rewritten as twice the sum over ordered pairs with $i < j$:

$$\sum_{i\neq j} a_i a_j x_i x_j = 2 \sum_{1\leq i<j\leq N} a_i a_j x_i x_j. \tag{8}$$

Substituting this back into the expansion yields the Hamiltonian $H(x)$:

$$H(x) = \sum_{i=1}^{N} (a_i^2 - 2Ta_i)x_i + 2 \sum_{1\leq i<j\leq N} a_i a_j x_i x_j + T^2. \tag{9}$$

From this general form, we read off the QUBO matrix entries $Q_{i,j}$ and the constant offset c. The QUBO objective is to minimize $x^\top Q x + c$, where x is the vector of variables. The entries are given by:

$$Q_{i,i} = a_i^2 - 2Ta_i, \quad \forall i \in \{1, \ldots, N\}, \tag{10}$$

$$Q_{i,j} = 2a_i a_j, \quad \forall 1 \leq i < j \leq N, \tag{11}$$

$$c = T^2. \tag{12}$$

Note that $Q_{i,j} = Q_{j,i}$ for $i > j$ due to symmetry, and the off-diagonal terms for $i \neq j$ are captured by the factor of 2 in the summation.

Penalty Weight Justification. The Subset Sum Problem formulation above contains no constraints; any assignment of binary values to x_i corresponds to a valid subset. Consequently, no penalty terms are required, and no penalty weights λ are needed. The Hamiltonian consists solely of the objective function terms.

Correctness Argument. Minimizing $H(x)$ over $x \in \{0, 1\}^N$ recovers the optimal solution because $H(x)$ is exactly the squared deviation $(\sum a_i x_i - T)^2$ for any binary vector x. Since the squared deviation is non-negative and zero if and only if the subset sum equals the target, the global minimum of $H(x)$ corresponds to the subset with the minimum squared deviation. Because the square function preserves the ordering of non-negative values, the subset minimizing the squared deviation also minimizes the absolute deviation. Thus, the configuration x^* that minimizes $H(x)$ identifies the subset of integers whose sum is closest to T.

Illustrative Example. Consider an instance with $N = 3$ integers, values $a_1 = 1$, $a_2 = 2$, $a_3 = 3$, and target $T = 4$. The general formulas yield the following QUBO parameters:

$$Q_{1,1} = 1^2 - 2(4)(1) = -7, \tag{13}$$
$$Q_{2,2} = 2^2 - 2(4)(2) = -12, \tag{14}$$
$$Q_{3,3} = 3^2 - 2(4)(3) = -15, \tag{15}$$
$$Q_{1,2} = 2(1)(2) = 4, \tag{16}$$
$$Q_{1,3} = 2(1)(3) = 6, \tag{17}$$
$$Q_{2,3} = 2(2)(3) = 12, \tag{18}$$
$$c = 4^2 = 16. \tag{19}$$

The resulting Hamiltonian is:

$$H(x) = -7x_1 - 12x_2 - 15x_3 + 4x_1x_2 + 6x_1x_3 + 12x_2x_3 + 16. \tag{20}$$

Evaluating $H(x)$ for all $2^3 = 8$ configurations reveals the minimum energy. The configuration $x^* = (1, 0, 1)$ yields:

$$H(1, 0, 1) = -7(1) - 12(0) - 15(1) + 4(0) + 6(1) + 12(0) + 16 = -22 + 6 + 16 = 0. \tag{21}$$

This corresponds to selecting a_1 and a_3, giving a sum of $1 + 3 = 4$, which matches the target exactly with zero deviation. The minimum energy is 0, confirming the optimal subset.